



KLE Technological University

Creating Value,
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Dr. M. S. Sheshgiri Campus, Belagavi

Department of
Electronics and Communication Engineering

Minor-2 Project Report
on
**Automated Student Performance Evaluation
System.**

By:

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CERTIFICATE

This is to certify that project entitled “**Automated Student Performance Evaluation System**” is a bonafide work carried out by the student team of ”**Ashwini M Dambal [02FE22BEC406], Pranjal Shindolkar [02FE22BEC416], Roopali Karadiguddi [02FE22BEC417], Snehal Sambaragi [02FE22BEC421]**”. The project report has been approved as it satisfies the requirements with respect to the mini project work prescribed by the university curriculum for B.E. (VI Semester) in Department of Electronics and Communication Engineering of KLE Technological University Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2023-2024.

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-The project team B14

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ABSTRACT

This project aims to revolutionize the grading process within educational institutions by developing a system focused on efficiency and accuracy. The system is designed to extract student data from input Excel sheets, perform calculations to determine total marks and percentages, and generate a comprehensive results sheet with individual grade assignments. By automating these tasks, the system eliminates manual calculations and potential errors, thus expediting the grading process while ensuring consistency and fairness in assessment practices. The project seeks to provide a scalable and user-friendly solution tailored to the evolving needs of educators and administrators in today's educational landscape.

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Chapter 1

Introduction

In the ever-evolving landscape of education, the grading process remains a cornerstone of academic evaluation. However, traditional methods of calculating and recording grades are often time-consuming and prone to errors, leading to inefficiencies and inconsistencies. This project aims to revolutionize the grading process within educational institutions by developing an automated system focused on efficiency and accuracy. By leveraging the power of automation, the system extracts student data from input Excel sheets, performs necessary calculations to determine total marks and percentages, and generates a comprehensive results sheet with individual grade assignments. This not only eliminates the need for manual calculations but also minimizes potential errors, ensuring a fair and consistent assessment process. Our solution is designed to be scalable and user-friendly, catering to the dynamic requirements of educators and administrators in today's educational environment.

1.1 Motivation

Efficiency Improvement: Manual grading is a time-consuming process that can delay the availability of student results. Automating this process significantly reduces the time required to grade assignments and exams.

Error Reduction: Human errors in manual calculations can affect the accuracy of grades. An automated system ensures precise calculations, thereby enhancing the reliability of the grading process.

Consistency and Fairness: Automated systems apply the same criteria and calculations uniformly to all students, ensuring a fair and consistent grading process.

Scalability: As educational institutions grow, the volume of grading increases. A scalable system can handle large datasets efficiently, making it suitable for institutions of varying sizes.

User-Friendliness: A system designed with a user-friendly interface can be easily adopted by educators and administrators, reducing the learning curve associated with new technologies.

1.2 Objectives

The primary objective of this project is to develop an automated grading system that significantly enhances the efficiency, accuracy, and fairness of the grading process within educational institutions. The system is intended to:

Automate Data Extraction: Seamlessly extract student data from input Excel sheets to eliminate the need for manual data entry.

Perform Accurate Calculations: Automatically compute total marks and percentages for each student, ensuring precision and eliminating human error.

Generate Comprehensive Results: Produce detailed results sheets that include individual grade assignments, providing clear and comprehensive feedback to students and educators.

Ensure Consistency and Fairness: Apply uniform grading criteria and calculations to all students, ensuring a consistent and fair assessment process.

Enhance User Experience: Develop a user-friendly interface that can be easily adopted by educators and administrators, minimizing the learning curve and facilitating widespread adoption.

Support Scalability: Design the system to handle large datasets efficiently, making it suitable for educational institutions of various sizes and accommodating future growth.

Provide Data Insights: Generate valuable data and analytics on student performance to support informed decision-making and targeted academic interventions.

By achieving these objectives, the project aims to deliver a robust and scalable grading solution that addresses the evolving needs of modern educational environments, ultimately contributing to improved educational outcomes.

1.3 Literature survey

[1]Title- Student Performance Assessment and Prediction System using Machine Learning

Objective: Student Performance Analysis System is an emerging field and is very crucial to schools and universities in helping their students and professors. Most of the pre-existing methods are based only on past academic performance of students. This paper aims to develop models which can predict the student's performance and grades while keeping in mind other equally essential personality factors like interests, attributes and opinions (IAO variables) which affect their lifestyle. It uses various machine learning and deep learning techniques to predict the performance of students, and basic exploratory data analysis to derive various correlations of student's performance with psychographic attributes.

Methods:

- 1.Efficient partial product addition schemes: Wallace tree, Dadda tree, and compressors.
- 2.The Wallace tree method : It utilizes a tree structure to add partial products efficiently.

Limitations:

- 1.Data Collection and Model Training Limitations
- 2.End-to-End Platform Solution Limitations
- 3.Utility for Teachers Limitations.

[2]Title-Data Mining Analysis on Student's Academic Performance through Exploration of Student's Background and Social Activities

Objective: Educational data mining techniques are widely used in academic prediction on student performance in classroom education. However most of the existing researches were studied and evaluated student coursework performance against the passing grade of the exam. In this paper, we performed analysis to identify the significant and impact of student background, student social activities and student coursework achievement in predicting student academic performance. Supervised educational data mining techniques, namely Naïve Bayesian, Multilayer Perceptron, Decision Tree J48 and Random Forest were used in predicting mathematic performance in secondary school. The prediction was performed on 2-level classification and 5-level classification on final grade. The experimental results have shown that student background and student social activities were significant in predicting student performance on 2-level classification. The model can be used for early predicting student performance to help in improving student performance on the subject.

Methods:

- 1.Algorithms:Naive Bayesian,Multilayer Perceptron,Decision Tree (J48),Random Forest.
- 2.Programming Languages: Python, R and Weka.

Limitations:

1. Dependence on Accurate Data
2. Model Generalization
- 3.Potential for Bias
- 4.Privacy Concerns

[3]Title-Student Performance Evaluation in Web Based Access to Robot Supported Laboratories

Objective: To enhance distance education through the development of an intelligent interface for remote student access to robot-supported laboratories, enabling hands-on experimentation, performance evaluation, and adaptive coaching based on individual student knowledge and skill levels.

Methods:

- 1.Development of an intelligent interface for remote student access to robot-supported labs
- 2.Utilization of error quantization module for performance evaluation
- 3.Implementation of artificial neural network for error type classification
- 4.Inclusion of an authentication subsystem for security purposes

Limitations:

- 1.Potential inaccuracies in error quantization and classification.
- 2.Constraints in adaptability due to predefined experiment parameters.
- 3.Challenges in user authentication and recall.
- 4.Limitations in providing adequate assistance for diverse student skill levels.



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Figure 1.1: Logo of KLE Technological University

1.4 Problem statement

”AcademicPulse: Real-time Student Performance Analysis”

1.5 Application in Societal Context

1. Enhanced Efficiency: Educators often spend a substantial amount of time on administrative tasks, such as grading, which can detract from their primary role of teaching and mentoring students. Automating these tasks allows teachers to focus more on instructional activities and student engagement, thereby enhancing the overall quality of education.

2. Accuracy and Fairness: Manual grading is susceptible to human error, which can lead to inaccuracies and inconsistencies in student evaluations. An automated system ensures precise calculations and uniform application of grading criteria, promoting fairness and objectivity in the assessment process. This fosters a more equitable academic environment, where students are assessed solely on their performance.

3. Scalability: Educational institutions vary widely in size, from small schools to large universities. A scalable grading solution can accommodate the needs of any institution, regardless of its size. This flexibility makes the system a valuable tool for a broad spectrum of educational settings, from primary schools to higher education.

4. Data Management and Analytics: Automated grading systems can facilitate the collection and analysis of student performance data over time. This data can provide valuable insights into academic trends, helping educators identify areas where students may need additional support and enabling more informed decision-making.

5. Support for Diverse Learning Environments: As educational paradigms shift towards more diverse and inclusive learning environments, the need for adaptable and efficient administrative tools becomes increasingly important. An automated grading system supports this evolution by providing a reliable and consistent method of evaluation, regardless of the instructional model employed.

6. Reduction of Bias: Automation in grading can help mitigate unconscious biases that may affect manual grading. By relying on objective criteria and algorithms, the system ensures that all students are evaluated based on the same standards, thus promoting inclusivity and fairness.

1.6 Project Planning

1. Project Initiation

Scope: Automate grading to improve accuracy and reduce workload.

Deliverables: Software, documentation, training materials.

Stakeholders: Educators, administrators, IT staff, students.

Feasibility Study: Assess technical, operational, and financial aspects.

2. Requirements Gathering

Stakeholder Input: Interviews and surveys to understand needs.

Documentation: Functional and non-functional requirements.

3. System Design

Architecture: Define data flow and system components.

Design Specs: Create UI mockups, database schema, and algorithms for grade calculations.

4. Development Phase

Setup: Choose tools, configure version control

5. Core Development

Data Extraction: Read and validate Excel data.

Calculation Engine: Develop algorithms for marks and grades.

Report Generation: Design and implement report templates.

Testing: Unit, integration, and system testing; debug issues.

6. User Training and Documentation

User Manuals: Create installation guides and usage instructions.

Training: Conduct workshops for educators and administrators.

7. Deployment

Testing: Final testing in a staging environment.

Go Live: Deploy to production, monitor and address issues.

8. Maintenance and Support

Support: Provide technical assistance, establish helpdesk.

Maintenance: Regular updates, performance monitoring.

9. Evaluation and Continuous Improvement

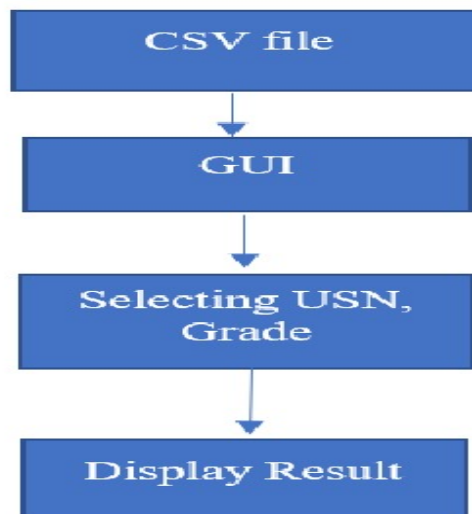
Feedback: Collect user feedback and performance data.

Enhancements: Implement improvements based on feedback and trends.

Chapter 2

System design

2.1 Block diagram



2.2 Block diagram Explanation

CSV file:

This block represents the data input for the system. The CSV file contains detailed records of student performance, including USNs (University Seat Numbers), scores in various subjects, attendance records, and possibly other metrics like participation or extracurricular activities. This file serves as the comprehensive data source for evaluating student performance.

GUI (Graphical User Interface):

The GUI is the interface through which users interact with the performance evaluation system. It allows users to upload the CSV file, navigate through different options, and perform various

actions such as selecting specific students or performance metrics. The GUI ensures the system is user-friendly and accessible for teachers, administrators, or other stakeholders.

Selecting USN, Grade:

In this block, users utilize the GUI to filter and select specific data points, such as a particular student's USN or a specific grade level. This step involves querying the CSV data to isolate the performance metrics of interest. Users can focus on individual students or groups based on criteria like grade level, subject performance, attendance, etc.

Display Result:

After making selections, the system processes the data to evaluate student performance. The results are then displayed on the GUI. This could include detailed performance reports, comparative analysis with peers, graphs showing progress over time, and overall evaluation summaries. The display provides actionable insights for educators to understand and improve student performance.

2.3 Design alternatives

Backend Processing with Python: Implementing a backend system using Python for data extraction, calculation, and report generation.

Web Application Interface: Creating a web-based interface for ease of access and usability among educators and administrators.

Machine Learning Integration: Incorporating machine learning algorithms for predictive analytics on student performance.

Cloud Integration: Utilizing cloud services for scalability and efficient data storage and processing.

Mobile Application: Developing a mobile app for on-the-go access to student performance data and analytics.

Chapter 3

Implementation details

Calculate Grade Function: Develop a function to compute weighted percentages and total percentages based on given marks (ISA1, ISA2, ESA, Assignments).

Assign Grade Algorithm: Define an algorithm to assign grades ('A+', 'A', 'B', 'C', 'X (Fail)') based on predetermined percentage thresholds.

Calculate Overall Percentage Function: Create a function to read data from a CSV file, calculate overall percentages for each student, and determine their grades. Optionally, filter results by a specific student's USN.

Error Handling: Implement mechanisms to handle potential errors like missing files (FileNotFoundError) or incorrect data formats (KeyError). Display appropriate error messages to users.

Display Results Algorithm: Develop a function to clear a display area and present results for individual students or lists of students in a clear and readable format.

Filtering by Division and Grade: Implement functionality to filter students based on selected division and grade criteria. Display filtered results accordingly.

Display All Students: Design a function to retrieve and display data for all students, utilizing the calculate overall percentage function to fetch comprehensive data.

Find Student by USN: Create a function to locate and display student information based on a provided USN (University Serial Number).

CSV File Selection: Incorporate functionality to allow users to select a CSV file containing student data for processing. Use a file dialog for user convenience.

GUI Setup: Utilize a graphical user interface (GUI) library such as Tkinter to build the main application window, input fields, buttons, and output display areas. Implement event handlers to link GUI components with corresponding backend functions, enabling user interaction and data processing.

3.1 Algorithm

Function: calculate grade(subject marks)

Process: Converts individual marks (ISA1, ISA2, ESA, Assignments) into weighted percentages and calculates the total percentage for a subject.

Function: assign grade(percentage)

Process: Assigns a grade ('A+', 'A', 'B', 'C', 'X (Fail)') based on predefined percentage thresholds.

Function: calculate overall percentage(file path, usn=None)

Process: Reads CSV data, calculates the overall percentage for each student, determines their grade, and handles filtering by USN if provided.

Error Handling: Catches FileNotFoundError and KeyError to handle missing files or data, displaying appropriate error messages using messagebox.

Function: display results(students)

Process: Clears the text widget and displays the results, handling both individual student data and lists of students.

Function: filter by grade()

Process: Filters students based on the selected division and grade, then displays the filtered results.

Function: display all students()

Process: Calls calculate overall percentage to get all student data and displays it.

Function: find student by usn()

Process: Retrieves and displays information for a student based on their USN.

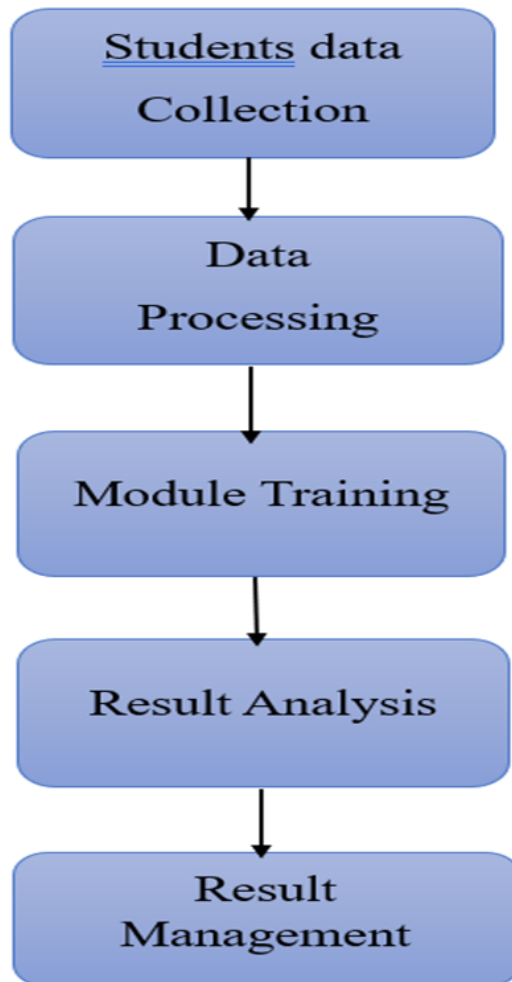
Function: select csv file()

Process: Opens a file dialog to select the CSV file and sets the file path for processing.

Components: Uses Tkinter to create the main window, entry fields, buttons, and a text widget for displaying results.

Event Loop: root.mainloop() runs the Tkinter event loop, enabling user interaction with the GUI

3.2 Flowchart



Chapter 4

Optimization

4.1 Introduction to Optimization

Optimization is a critical process in numerous disciplines, including mathematics, engineering, economics, and computer science. It involves systematically improving or finding the best solution among a set of feasible alternatives. The primary goal of optimization is to maximize benefits, minimize costs, or achieve other desired outcomes within given constraints.

At its essence, optimization revolves around formulating an objective function that quantifies the goal to be achieved (such as maximizing profit or minimizing travel time), defining decision variables that can be adjusted to influence this objective, and incorporating constraints that limit the feasible solutions.

There are various types of optimization problems, each suited to different complexities and scenarios. These include linear programming for problems with linear relationships, nonlinear optimization for more complex relationships, discrete optimization for problems with discrete decision variables, and multi-objective optimization for balancing competing goals.

Optimization algorithms and techniques play a crucial role in helping organizations and individuals make informed decisions. By applying these methods, they can improve efficiency, innovate processes, allocate resources effectively, and ultimately enhance competitiveness in their respective fields. The application of optimization spans from operational planning and logistics to financial modeling, engineering design, and beyond, making it a cornerstone of modern problem-solving and decision-making strategies.

4.2 Optimization of project

Algorithmic Efficiency:

Optimize algorithms for calculating grades and percentages to ensure fast and efficient processing, especially when dealing with large datasets.

Database Management:

Implement efficient data storage and retrieval mechanisms to minimize latency and enhance system responsiveness.

User Interface (UI) Enhancements:

Conduct usability studies to refine the GUI, making it intuitive and user-friendly. Incorporate user feedback to improve navigation and data presentation.

Error Handling and Logging:

Enhance error handling mechanisms to provide informative error messages and log critical events for troubleshooting and system maintenance.

Security Measures:

Implement robust security protocols to protect student data and ensure compliance with data privacy regulations (e.g., GDPR, HIPAA).

Automated Testing and Continuous Integration (CI/CD):

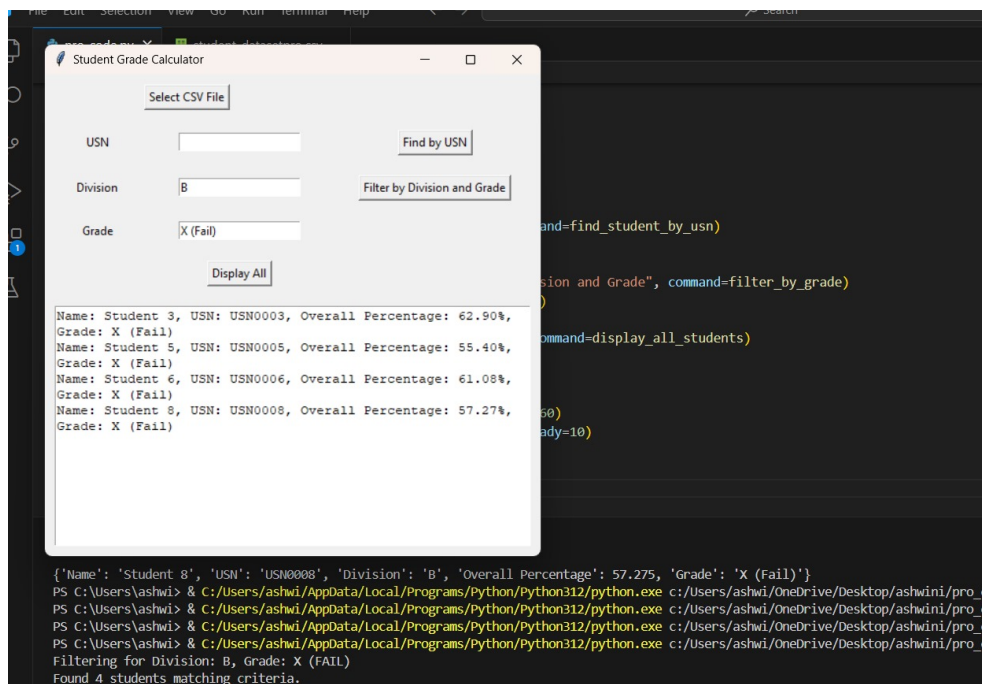
Establish automated testing pipelines to validate system updates and improvements, ensuring stability and reliability across releases.

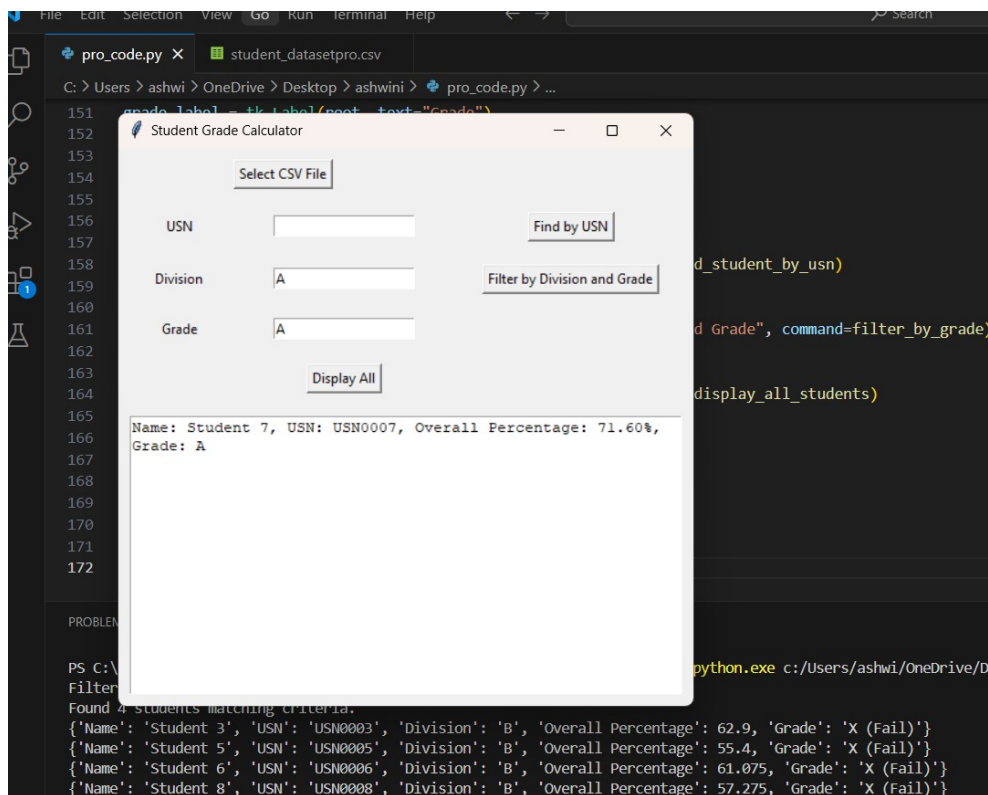
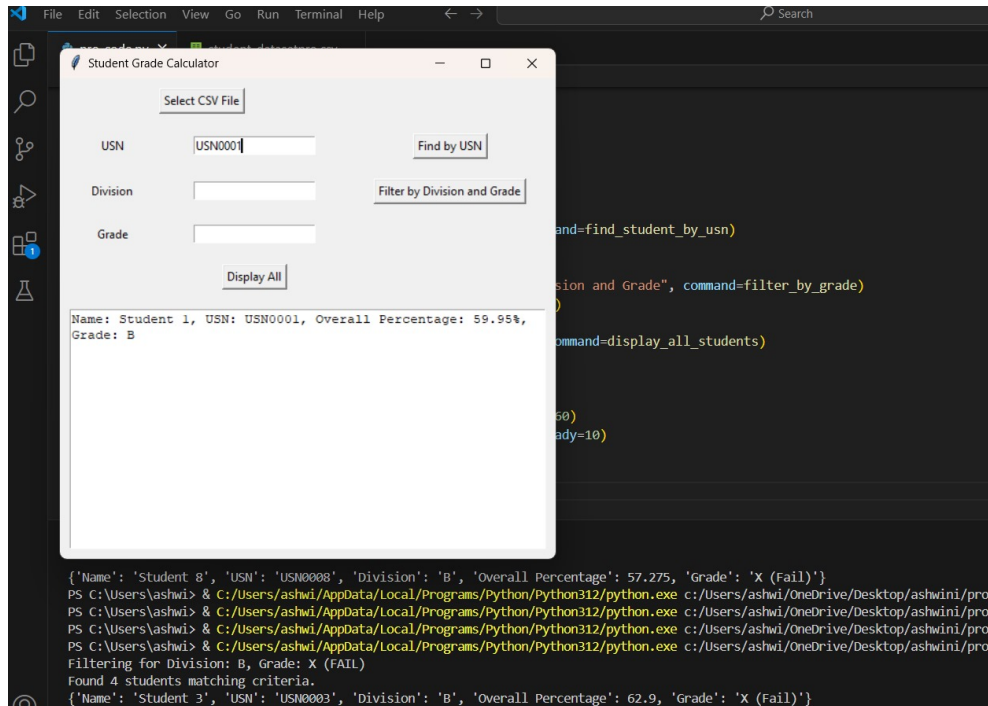
Feedback and Iterative Development:

Regularly solicit feedback from users (educators, administrators, students) to identify pain points and opportunities for enhancement. Adopt an iterative development approach to continuously improve the system.

Chapter 5

Results





Chapter 6

Conclusions and future scope

6.1 Conclusion

In conclusion, our project introduces an automated student performance evaluation system using machine learning, streamlining grading processes and ensuring consistency. With Python and key libraries, we've created a user-friendly interface for inputting data and viewing results. By prioritizing data security and compliance, we provide a robust solution for educators to enhance student assessment. This project signifies a step forward in modernizing educational assessment practices, with potential for further improvement and innovation in the future.

6.2 Future scope

Integration with Learning Management Systems (LMS):

Integrate the automated evaluation system with existing LMS platforms to streamline data exchange and enhance user experience for educators and administrators.

Machine Learning for Predictive Analysis:

Implement machine learning models to predict student performance trends based on historical data. This could provide insights for early intervention and personalized learning strategies.

Enhanced Visualization and Reporting:

Develop interactive data visualization tools to present evaluation results graphically. This can include trends over time, comparative analysis, and performance metrics for individual students and cohorts.

Real-time Data Analytics:

Enable real-time analytics capabilities to monitor student progress dynamically. This can include alerts for significant changes in performance or attendance patterns.

Multi-dimensional Grading Criteria:

Extend grading criteria beyond academic performance to include co-curricular activities, behavioral assessments, and soft skills development.

Scalability and Cloud Integration:

Optimize the system architecture for scalability, allowing it to handle larger datasets and support growing educational institutions. Integrate with cloud services for improved accessibility and storage.

Feedback Mechanisms and Student Portals:

Develop features for students to access their performance reports, receive feedback from educators, and track their academic journey over time.

Mobile Application Development:

Create a mobile application version of the system for convenient access on smartphones and tablets, enhancing usability for students, parents, and faculty.

Chapter 7

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